

SYED SAHAJADA MAHAFUJUL ALAM, PhD

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Computational biologist specializing in structure-based drug discovery, molecular docking and dynamics, immunoinformatics, and host-pathogen genomics/metagenomics. Author of 16 peer-reviewed publications and 6 book chapters in computational biology and cancer immunotherapy (279 citations, h-index 9). Experienced in Python-based bioinformatics workflow automation and HPC-based analysis.

CURRENT POSITION

Visiting Researcher, Indian Institute of Science Education and Research (IISER) Kolkata 2026–present

EDUCATION

PhD, Biological Sciences - Aliah University, Kolkata / Thesis: *Structural Insights into PD-L1 Inhibition in Cancer Immunotherapy* 2020–2026

MSc, Zoology - University of Calcutta 2017–2019

BSc, Zoology - University of Calcutta 2014–2017

RESEARCH EXPERIENCE

Visiting Researcher, IISER Kolkata 2026–present

- Expanded into host-pathogen interaction research: bacterial/mammalian cell culture, Nanopore whole-genome sequencing, comparative genomics, and metagenomics.

Senior Research Fellow, Applied Biochemistry Lab, Aliah University 2023–2026

- Led computational studies on PD-L1 structure/function; identified small-molecule and peptide inhibitors via docking, MD simulations, and virtual screening; designed a multi-epitope cancer vaccine.

Junior Research Fellow, Applied Biochemistry Lab, Aliah University 2020–2023

- Investigated nsSNP impacts on CD274/PD-L1 and viral (SARS-CoV-2, ACE2) protein interactions using in silico approaches.

SELECTED PUBLICATIONS

- Alam, S.S.M. et al. (2026). Structure-guided identification of phytochemical inhibitors targeting PD-L1. *ChemistrySelect*, 11(1).
- Alam, S.S.M. et al. (2025). Immunoinformatics-based design of a multi-epitope cancer vaccine targeting PD-L1. *Scientific Reports*, 15(1).
- Alam, S.S.M. et al. (2025). Structural insights into nsSNP impacts in CD274 on PD-1/PD-L1 interaction. *Biochemical and Biophysical Research Communications*, 152679.
- Gorain, C., ...Alam, S.S.M. et al. (2024). Functional characterization of ligand-binding domains of JlpA in *C. jejuni*. *International Journal of Biological Macromolecules*, 264.
- Alam, S.S.M. et al. (2023). Tanshinone IIA targeting cell signaling pathways in cancer therapy. *Pharmacological Reports*, 75(4).

TECHNICAL SKILLS

- Structure-Based Drug Discovery:** AutoDock/Vina, PyRx, PyMOL, UCSF Chimera, Discovery Studio; GROMACS for MD simulations
- Genomics & Metagenomics:** Oxford Nanopore sequencing, genome assembly/annotation, AMR & virulence gene analysis, comparative genomics
- Programming & Automation:** Python, Selenium/Playwright/BeautifulSoup, Linux/Bash, HPC & GPU-accelerated workflows
- Wet Lab:** PCR, gel electrophoresis, SDS-PAGE, Western blotting, bacterial/mammalian cell culture, microbiology

FELLOWSHIPS & AWARDS

- CSIR-UGC NET Fellowship, Government of India (2021) | DBT-JRF Fellowship, Government of India (2020, 2021)
- Best Oral Presentation Award — ETBR Conference, Aliah University (2025)